

Title

MGT 3745B Business Programming: Real-Time Web App Integrations with Google Firebase

Author

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Grade/Grade Span

College/University Undergraduates

Summary**Standards**

- Georgia Institute of Technology: Scheller College of Business
 - Information Technology Management (ITM) Concentration & Certificate
- Google Firebase

Information Technology Management (ITM) Concentration & Certificate

The Information Technology Management (ITM) certificate equips graduates to connect business challenges with information technology solutions. It is also referred to as MIS (Management Information Systems), BIS (Business Information Systems), or IS (Information Systems).

The ITM Certificate emphasizes designing and developing computer-based business information systems and their importance in organizations. It includes topics like database design, systems analysis, data communications, and various

concepts for integrating information technology into business functions. Google Firebase Students will learn how Firebase's Realtime Database addresses complex business challenges, such as managing inventory, enhancing customer relationship management (CRM), creating recommendation engines, and delivering personalized user experiences. The Firebase Realtime Database provides secure, direct access from client-side JavaScript applications. Students will understand how data is persistently stored, allowing applications to remain responsive even offline. Upon reconnection, Firebase automatically synchronizes local and remote changes with conflict resolution. Students will configure Firebase in web apps through project-based learning, establish real-time two-way data bindings, and simulate business scenarios where immediate data updates enhance operational efficiency and customer engagement. The course will focus on semi-structured data modeling, cloud service integration, event-driven updates, and best practices for maintaining responsive, resilient applications. 5Es	
Engage. Students will begin the unit by observing a live demo of a basic business web app showing real-time updates using Firebase. The instructor will manipulate inventory items or customer profiles in the database, and students will	How might understanding the tangible benefits of real-time applications in solving business problems contribute to students' initial self-efficacy in potentially learning to work with such technologies later in the unit

watch instant, automatic updates reflected on the user interface without refreshing the page.

A guided discussion will follow:

- Why is real-time synchronization critical for modern businesses?
- What customer expectations exist for instantaneous interactions in apps?
- What types of businesses could most benefit from this capability?

Students will brainstorm industries (e.g., e-commerce, hospitality, healthcare) where these capabilities could significantly solve operational problems.

Explore. Students will be assigned to small project teams and provided with **JavaScript starter code templates** connected to a Firebase project. Each team will receive a fictional business profile (such as a boutique store, a restaurant, or a personal coaching service) and an initial database schema.

Teams will explore:

- **Reading and writing to the Firebase Realtime Database**
- Creating listeners that automatically update the user interface when changes occur
- Simulating offline data changes and observing conflict resolution and synchronization

How does the guided discussion build upon students' prior knowledge of business operations and customer interactions, and are there specific questions planned to elicit an existing understanding of synchronization challenges in different business contexts?

Does the discussion effectively tap into students' intrinsic motivation by highlighting the potential impact on various industries?

Is the starter code broken into components that minimizes extraneous cognitive load during this initial exploration phase to manage the intrinsic cognitive load associated with learning new technologies like Firebase?

Will students be capable of using practical debugging strategies or resources when encountering unexpected behavior or errors? How can I assert?

Students will document observations:

- How does the database behave offline and in reconnect states?
- How does real-time data updating enhance business responsiveness?

Explain. The instructor will present mini-lectures and live coding walkthroughs covering:

- Principles of NoSQL and semi-structured data design in Firebase
- Authentication and security rules in Realtime Database
- How offline data persistence and multi-user synchronization work under the hood
- Trade-offs between relational (SQL) vs NoSQL (document-oriented) databases for real-time needs

Students will individually complete knowledge checks with short coding exercises to reinforce:

- How to create, read, update, and delete records securely
- How to handle edge cases like conflicting edits
- How to structure data hierarchies for efficient reads and writes

Elaborate. Students will extend their projects by building a feature that demonstrates a business-oriented wicked problem solution, such as:

- A real-time low-stock alert system for inventory
- A collaborative CRM tool where multiple employees can edit customer notes simultaneously

Will the requirement to document observations promote metacognition and self-assessment as students reflect on their learning process and the behavior of the system?

How do the live coding walkthroughs exemplify the worked-example effect, providing step-by-step solutions that students can learn from?

Can students be encouraged to view errors in their coding exercises as learning opportunities, rather than as failures that might negatively impact self-efficacy?

What insights or approaches did students learn from observing or critiquing peers' implementations that influenced their approach?

• A live chat feature with message persistence across user sessions • A dynamic personalized recommendation list based on browsing behavior stored in the database Students will be expected to: • Model and store appropriate data for their new feature • Create event-driven updates to reflect changes on the frontend immediately • Justify design choices (data structure, security rules, user experience enhancements) Peer critiques will allow students to test each other's apps and give feedback on functionality, usability, and scalability. Evaluate. Each student will complete a portfolio submission including: • A working web app demonstrating Firebase integration • A diagram showing their database structure • A short reflective essay answering: ○ How does real-time data synchronization enhance business operations and customer experience? ○ What challenges did they face implementing real-time updates? ○ How could this model scale for a large enterprise application? Submissions will be evaluated based on a rubric assessing technical execution, database design, innovative use of real-time capabilities, and critical reflection.	Will there be an opportunity to observe any confidence changes in students from the start to the completion of this activity? How are the peer critiques designed to foster collaborative learning and observational learning? How does the level of technical execution in the web app reflect the modeling of secure coding practices and efficient database interactions demonstrated by the instructor during the "Explain" phase To what extent does their understanding align with the potential for real-time data to provide relevant and meaningful experiences for users, thereby potentially increasing motivation and engagement
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